

Software Requirements Specification (SRS)

Project Title:

ShadowLight : *Illuminating Every Step on the Web*

An AI-Powered Chrome Extension for Accessibility and Guided Web Interaction

Organization: FlairMinds Software Pvt. Ltd.

Version: 2.0

1. Introduction

1.1 Purpose

The purpose of this document is to define the software requirements for ShadowLight, a Chrome browser extension being developed by interns at FlairMinds. This document serves as a reference for developers, mentors, and stakeholders to understand the system's functionality, scope, and constraints.

1.2 Product Overview

ShadowLight is an AI-powered chatbot integrated into the Chrome browser that helps users understand websites, navigate web pages, and complete tasks easily. The extension focuses on improving accessibility, usability, and user guidance without modifying the original website code.

1.3 Intended Audience

This document is intended for:

- Development team members
- Project mentors and reviewers
- Web users

2. Overall Description

2.1 Product Perspective

ShadowLight operates as a browser extension and runs independently of the websites it interacts with. It overlays functionality on top of existing web pages, providing guidance, summaries, and accessibility improvements in real time.

2.2 Product Functions

The major functions of the system include:

- Website content summarization
- Step-by-step task guidance
- Color Contrast

2.3 User Classes and Characteristics

- **General Users:** Users who want quick understanding and easier navigation of websites.
- **Users with Accessibility Needs:** Users with visual challenges.
- **First-time Users:** Users unfamiliar with complex websites or workflows

No advanced technical knowledge is required to use the extension.

3. Objectives

The objectives of ShadowLight are:

- To summarize website content for faster understanding
- To enhance accessibility for users with disabilities
- To guide users step-by-step through common web tasks
- To improve overall usability and user experience

4. Scope

4.1 In-Scope

- **Chrome extension development:** Design and implementation of the solution as a Google Chrome browser extension.
- **AI chatbot interface:** An AI-powered chatbot available on any website to assist users with queries and actions.
- **Website summarization feature:** Provides concise summaries of the current webpage for quick understanding.
- **Accessibility enhancements:**
 - **Color Contrast:** Improves text and background contrast to enhance readability for users with visual impairments.
- **Task-based visual guidance system:** Guides users step-by-step by dimming the screen and highlighting relevant UI elements to complete tasks easily.

4.2 Out-of-Scope

- Complex backend server development
- Permanent modification of website source code
- Full automation of tasks without user interaction
- Deep integration with third-party internal systems

5. Functional Requirements

5.1 Chatbot Interface

- The system shall provide a chatbot interface accessible on any website.
- The chatbot shall accept natural language user commands.

5.2 Website Summarization

- The system shall summarize the content of the active webpage.
- The summary shall be concise and easy to understand.

5.3 Accessibility Features

- The system shall allow users to adjust contrast.

5.4 Task Guidance System

- The system shall support guided task execution (e.g., login).
- The system shall dim the background and highlight relevant UI elements.
- The system shall guide users step-by-step using “Next” and “Previous” controls.

6. Non-Functional Requirements

6.1 Performance

- The extension shall be lightweight and fast.
- User actions shall have minimal response time.

6.2 Compatibility

- The extension shall work on most modern websites.
- The extension shall support the latest versions of Google Chrome.

6.3 Security and Privacy

- The system shall not store unnecessary user data.
- User interactions shall remain private and secure.

7. Assumptions and Constraints

7.1 Assumptions

- Users have access to a modern Chrome browser.
- Websites follow standard HTML structures.
- Users grant required browser permissions.

7.2 Constraints

- Browser security policies may restrict access to some elements.
- Website structure differences may affect guidance accuracy.
- Dynamic or protected content may limit interaction.

8. Future Enhancements

- **Multi-language support:** Allows users to interact with the extension in different languages for better understanding and wider usability.
- **Voice-based interaction:** Enables users to give commands and receive responses using voice, reducing the need for manual interaction.
- **Personalized accessibility profiles:** Lets users save and apply their preferred accessibility settings across websites automatically.
- **Support for additional browsers:** Extends compatibility beyond Chrome to other browsers such as Firefox and Edge.